Experiment: Characterization of an RF Mixer

Objective

To characterize an RF mixer by applying RF and LO signals, observing the intermediate-frequency (IF) output, and determining the conversion loss.

Components Required

RF signal generator, LO signal generator, RF mixer, spectrum analyzer/power meter, RF cables and connectors, and suitable 50 Ω termination/load.

Theory

A mixer is a nonlinear RF device used for frequency conversion. When an RF signal of frequency f_{RF} and a local oscillator signal of frequency f_{LO} are applied, an IF component is produced mainly at the difference frequency:

$$\mathbf{f_{IF} = |f_{RF} - f_{LO}|}$$

Other mixing products, including the sum frequency, may also be present. Conversion loss is the difference between the applied RF input power and the measured IF output power:

$$\mathbf{\text{Conversion Loss} = P_{RF} - P_{IF}}$$

Experimental Setup

The RF signal generator was connected to the RF port of the mixer and the LO signal generator was connected to the LO port. The IF/output port was connected to the measuring instrument. The RF input power was maintained at approximately –3.00 dBm and the LO frequency was maintained at 10 MHz while the RF frequency was varied.

Observation & Graphs

f_RF (MHz)	f_LO (MHz)	Expected f_IF (MHz)	Measured f_IF (MHz)	P_IF (dBm)	Conversion Loss (dB)
30	10	20	19.99	-13.35	10.35
30	10	40	39.99	-13.37	10.37
40	10	50	49.97	-13.30	10.30
40	10	30	29.97	-13.50	10.50
50	10	60	59.99	-13.30	10.30
50	10	40	39.99	-13.20	10.20

The measured IF frequencies closely follow the expected difference-frequency relationship.

Calculation & Results

Given RF input power:
P_{RF} = –3.00 dBm

For f_{RF} = 30 MHz and f_{LO} = 10 MHz:
f_{IF} = |30 – 10| = **20 MHz**
Measured f_{IF} = **19.99 MHz**
Conversion Loss = –3.00 – (–13.35) = **10.35 dB**

For the other observations, the calculated conversion losses are **10.37 dB, 10.30 dB, 10.50 dB, 10.30 dB and 10.20 dB**, respectively.

Conclusion (Key Point / Takeaway)

The RF mixer was successfully characterized. The measured IF frequencies closely matched the expected difference frequencies, confirming frequency conversion by the mixer. The observed conversion loss was approximately 10.2–10.5 dB for the recorded measurements.